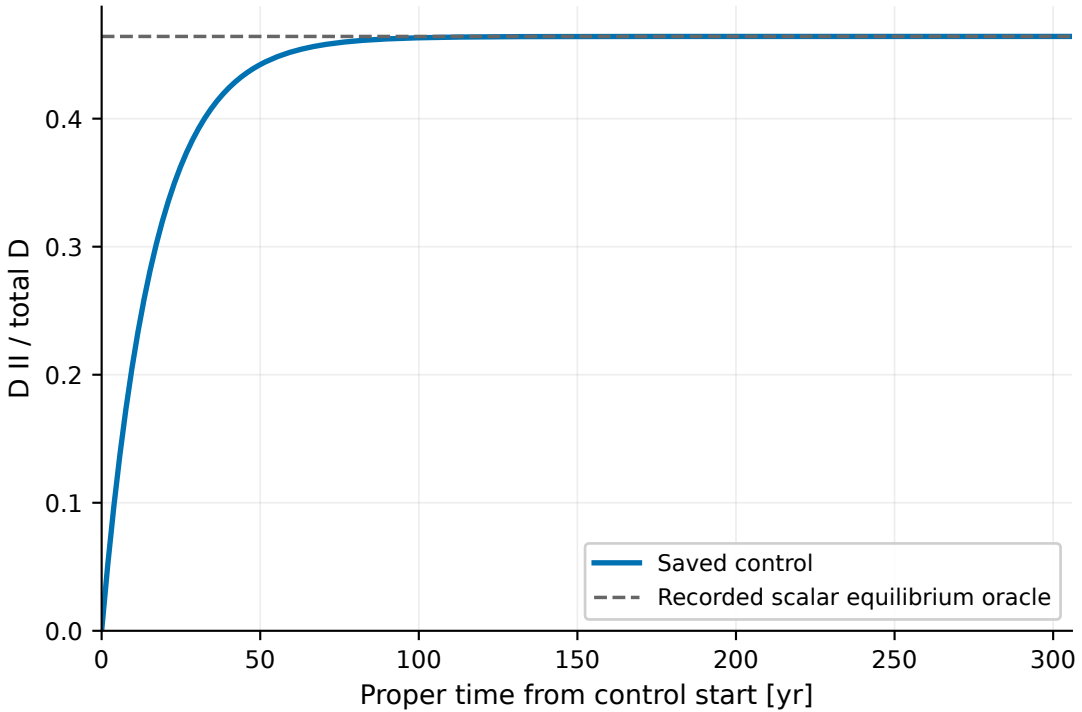
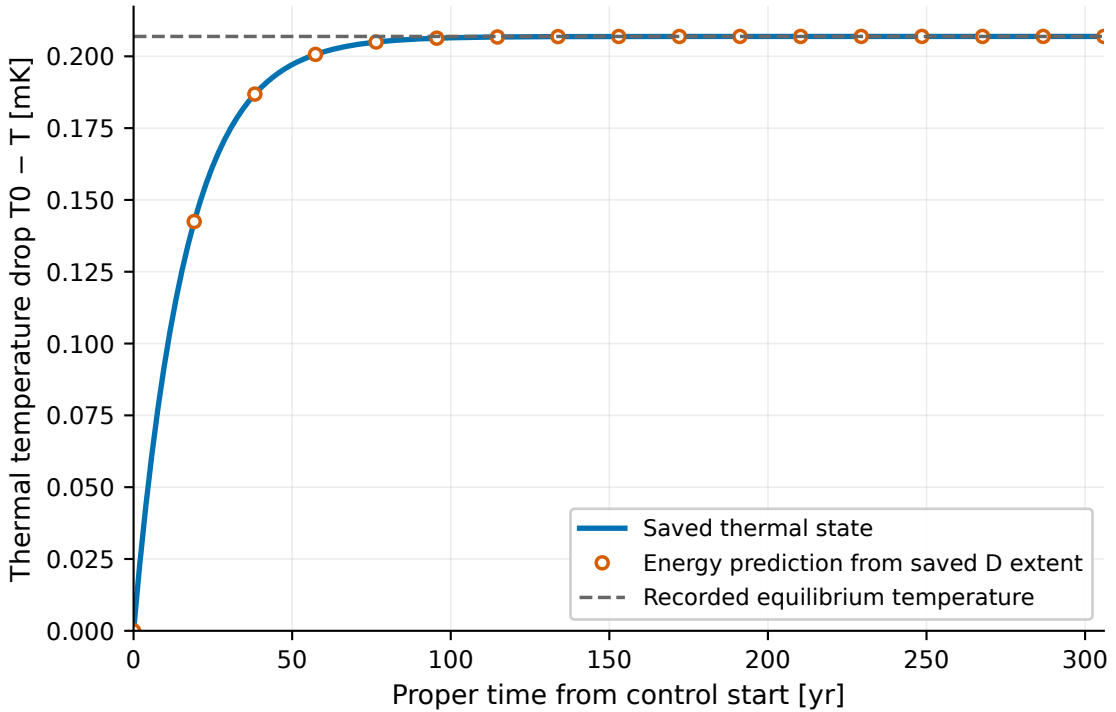
C02: isolated D/H charge exchange and its energy defect

Fixed-density closed gas: nH = 1 cm⁻³, T0 = 300 K; ground-state D/H exchange only.





Final D-ion fraction distance to the recorded equilibrium = 3.635e-09; final temperature drop = 0.206919573 mK.

The equilibrium reference comes from the saved scalar DB oracle; markers use saved reaction extent. Finite-time and sampled agreement are not continuum certificates.